TEST - verification

DB-POOL

Customer	TEST
Job number	TEST-001
Issued under	iec-default
Design rules	software defaults, not confirmed by the company's engineers

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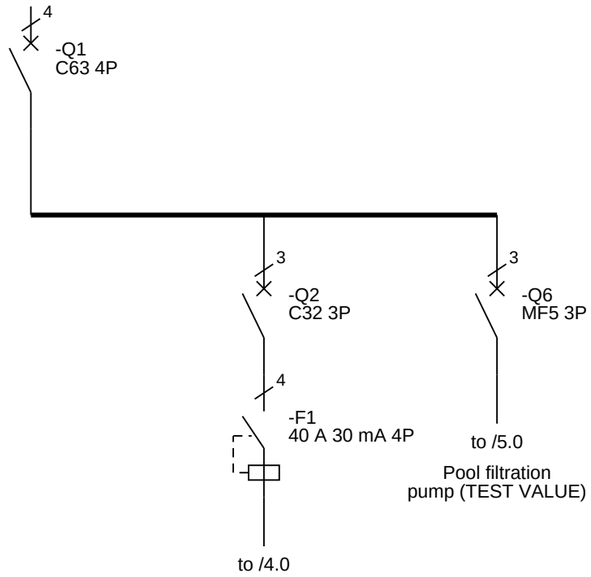
3/N/PE ~ 50 Hz
400 V TN-S



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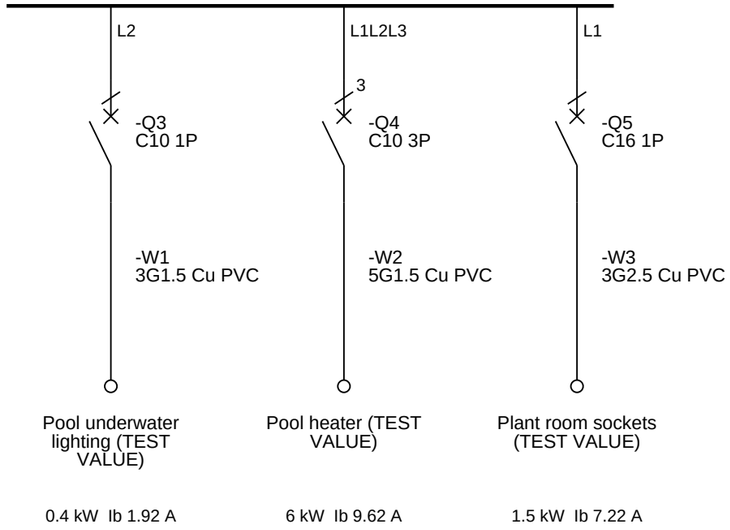
from -F1 40 A 30 mA 4P /3.1



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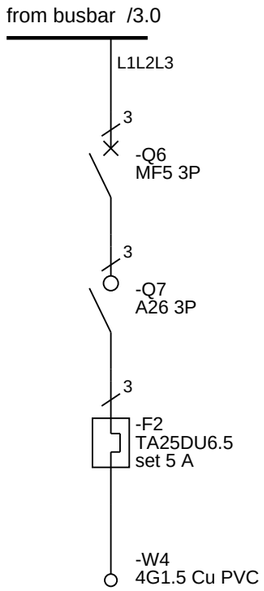

Pool filtration pump
(TEST VALUE)

2.2 kW Ib 5 A

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No.	Note
1	Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination.
2	Pool filtration pump (TEST VALUE): Ir 5 A, the typical Ir of a 2.2 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
3	Pool filtration pump (TEST VALUE): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2.
4	Phase imbalance 33.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
5	Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
6	No fault level given: every breaker's breaking capacity is left to be confirmed.
7	Leave 1 spare outgoing ways (20 %).
8	Loads without a power factor were taken at cos phi 0.9.
9	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
10	No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze.
11	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
12	Circuit rules are this software's defaults, not confirmed by the company's engineers.
13	Enclosure: none named, since 3 devices have no width (rail_widths_mm in the company settings).

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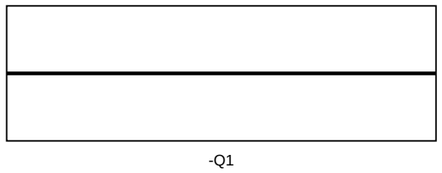
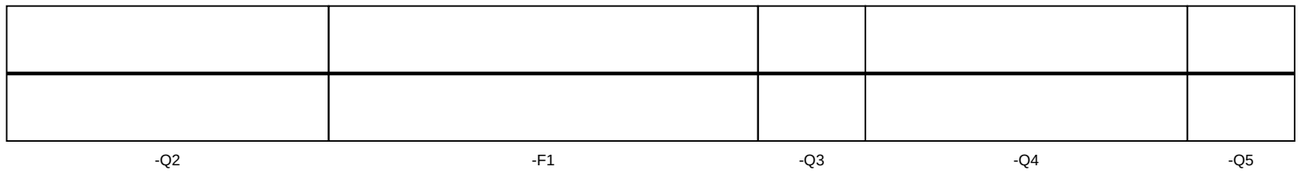
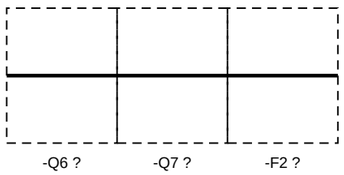
Incomer: 70 mm of rail



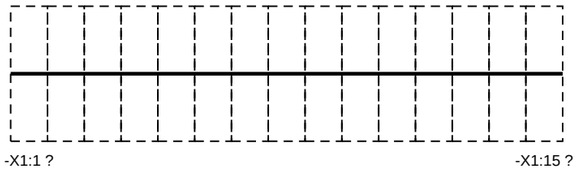
Residual current group: 210 mm of rail



Busbar: 0 mm of rail; 3 without a width



Terminals: 0 mm of rail; 15 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

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Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L2	Pool underwater lighting (TEST VALUE)	-W1	1.5	8WH1000-0AF00
-X1:2	N	Pool underwater lighting (TEST VALUE)	-W1	1.5	8WH1000-0AF00
-X1:3	PE	Pool underwater lighting (TEST VALUE)	-W1	1.5	8WH1000-0CF07
-X1:4	L1	Pool heater (TEST VALUE)	-W2	1.5	8WH1000-0AF00
-X1:5	L2	Pool heater (TEST VALUE)	-W2	1.5	8WH1000-0AF00
-X1:6	L3	Pool heater (TEST VALUE)	-W2	1.5	8WH1000-0AF00
-X1:7	N	Pool heater (TEST VALUE)	-W2	1.5	8WH1000-0AF00
-X1:8	PE	Pool heater (TEST VALUE)	-W2	1.5	8WH1000-0CF07
-X1:9	L1	Plant room sockets (TEST VALUE)	-W3	2.5	8WH1000-0AF00
-X1:10	N	Plant room sockets (TEST VALUE)	-W3	2.5	8WH1000-0AF00
-X1:11	PE	Plant room sockets (TEST VALUE)	-W3	2.5	8WH1000-0CF07
-X1:12	U	Pool filtration pump (TEST VALUE)	-W4	1.5	8WH1000-0AF00
-X1:13	V	Pool filtration pump (TEST VALUE)	-W4	1.5	8WH1000-0AF00
-X1:14	W	Pool filtration pump (TEST VALUE)	-W4	1.5	8WH1000-0AF00
-X1:15	PE	Pool filtration pump (TEST VALUE)	-W4	1.5	8WH1000-0CF07

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Cable	From	To	Type	Length
DB-POOL -W1	DB-POOL -Q3	Pool underwater lighting (TEST VALUE)	3G1.5 Cu PVC	20 m
DB-POOL -W2	DB-POOL -Q4	Pool heater (TEST VALUE)	5G1.5 Cu PVC	15 m
DB-POOL -W3	DB-POOL -Q5	Plant room sockets (TEST VALUE)	3G2.5 Cu PVC	10 m
DB-POOL -W4	DB-POOL -Q6	Pool filtration pump (TEST VALUE)	4G1.5 Cu PVC	15 m

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Qty	Description	Article	Order number	Devices
1	Circuit-breaker C63 4P	not selected		-Q1
1	Circuit-breaker C32 3P	not selected		-Q2
1	Residual current circuit-breaker 40 A 30 mA 4P	not selected		-F1
1	Circuit-breaker MF5 3P	ABB T2S160 MF 5		-Q6
1	Contactor A26 3P	ABB A26		-Q7
1	Overload relay TA25DU6.5 set 5 A	ABB TA25DU6.5		-F2
1	Circuit-breaker C10 1P	not selected		-Q3
1	Circuit-breaker C10 3P	not selected		-Q4
1	Circuit-breaker C16 1P	not selected		-Q5
11	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	DB-POOL -X1
4	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	DB-POOL -X1

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